Assignment – Classification GridSearchCV Algorithms

Dataset File Name: **CKD.csv**

1. **Identify your problem statement:**

   3 – Stages:

   Stage -1: Domain → Machine Learning
   Stage -2: Learning → Supervised Learning
   Stage -3: Learning → Supervised Learning (Classification)

2. **Tell basic info about the dataset:**

   Total Number of rows: 399
   Total Number of Col: 28

3. **Mention the pre-processing method:**

   Get Dummies:

   String to Number – Ordinal data – Mapping – Label Encoder

Classification (Categorical)	Classification (NUMERICAL)
No	0
Yes	1

Standard Scalar:

```
from sklearn.preprocessing import StandardScaler
```

```
df4_num = df[['loan_amount','rate']]
df4_num.head()
```

client_id	loan_amount	rate
46109	13672	2.15
46109	9794	1.25
46109	12734	0.68
46109	12518	1.24
46109	14049	3.13

```
ss = StandardScaler()

scaled_x = ss.fit_transform(df4_num)
scaled_x
```

```
array([[ 1.36502962e+00, -4.45676908e-01],
       [ 4.34647423e-01, -8.21544457e-01],
       [ 1.13991132e+00, -1.05959300e+00],
       [ 1.08817014e+00, -8.25720763e-01],
       [ 1.45547678e+00, -3.63989115e-02],
       [-2.51263533e-01, -5.33379336e-01],
       [-4.33117505e-01,  2.61555546e+00],
       [ 1.12127812e+00,  3.85408004e-01],
       [ 7.38856969e-01,  9.45033020e-01],
       [-9.43412224e-01,  4.35523677e-01],
       [-1.52688038e+00,  1.36683993e+00],
       [ 1.01648330e-01, -1.13476741e+00],
       [ 2.57831622e-01, -9.84420394e-01],
```

4. Develop a good model with good evaluation metric:

- To identify the best-performing model for the CKD.csv dataset using the following machine learning classification algorithms:

1. Support Vector Classification
2. Decision Tree
3. Random Forest
4. Logistic Regression
5. K Nearest Neighbor
6. Naïve Bayes
7. Adaboost Classification
8. XG Boost Classification
9. LG Boost Classification

Support Vector Classification:

F1_Score: 0.9924946382275899

The confusion Matrix:

```
[[51  0]
 [ 1 81]]
```

Classification Report:

The report:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	51
1	1.00	0.99	0.99	82
accuracy			0.99	133
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133

roc_auc _score: 1.0

Decision Tree:

F1_Score: 0.9700283472213296

```
The confusion Matrix:
 [[50  1]
 [ 3 79]]
```

Classification Report:

```
The report:
              precision    recall  f1-score   support

           0       0.94      0.98      0.96        51
           1       0.99      0.96      0.98        82

    accuracy                           0.97       133
   macro avg       0.97      0.97      0.97       133
weighted avg       0.97      0.97      0.97       133
```

roc_auc _score: 0.9719033955045432

Random Forest:

F1_Score: 0.9924667654735397

```
The confusion Matrix:
 [[50  1]
 [ 0 82]]
```

Classification Report:

```
The report:
              precision    recall  f1-score   support

           0       1.00      0.98      0.99        51
           1       0.99      1.00      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

roc_auc _score: 0.9997608799617408

Logistic Regression:

F1_Score: 0.9924946382275899

The confusion Matrix:
[[51 0]
[1 81]]

Classification Report:

The report:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	51
1	1.00	0.99	0.99	82
accuracy			0.99	133
macro avg	0.99	0.99	0.99	133
weighted avg	0.99	0.99	0.99	133

roc_auc _score: 1.0

K-Nearest Neighbor:

F1_Score: 0.9701163285572423

The confusion Matrix:
[[51 0]
[4 78]]

Classification Report:

The report:

	precision	recall	f1-score	support
0	0.93	1.00	0.96	51
1	1.00	0.95	0.97	82
accuracy			0.97	133
macro avg	0.96	0.98	0.97	133
weighted avg	0.97	0.97	0.97	133

roc_auc _score: 0.9873266379722621

Naïve Bayes (Gaussian) :

F1_Score: 0.9850141736106648

```
The confusion Matrix:
 [[51  0]
 [ 2 80]]
```

Classification Report:

```
The report:
              precision    recall  f1-score   support

           0       0.96      1.00      0.98        51
           1       1.00      0.98      0.99        82

    accuracy                           0.98       133
   macro avg       0.98      0.99      0.98       133
weighted avg       0.99      0.98      0.99       133
```

roc_auc _score: 1.0

AdaBoost Classification:

F1_Score: 0.9850141736106648

```
The confusion Matrix:
 [[51  0]
 [ 2 80]]
```

Classification Report:

```
The report:
              precision    recall  f1-score   support

           0       0.96      1.00      0.98        51
           1       1.00      0.98      0.99        82

    accuracy                           0.98       133
   macro avg       0.98      0.99      0.98       133
weighted avg       0.99      0.98      0.99       133
```

roc_auc _score: 1.0

XGBoost Classification:

F1_Score: 0.47029200909249863

```
The confusion Matrix:
 [[ 0 51]
 [ 0 82]]
```

Classification Report:

The report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	51
1	0.62	1.00	0.76	82
accuracy			0.62	133
macro avg	0.31	0.50	0.38	133
weighted avg	0.38	0.62	0.47	133

roc_auc _score: 0.9817073170731707

LGBoost Classification:

F1_Score: 0.47029200909249863

```
The confusion Matrix:
 [[ 0 51]
 [ 0 82]]
```

Classification Report:

The report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	51
1	0.62	1.00	0.76	82
accuracy			0.62	133
macro avg	0.31	0.50	0.38	133
weighted avg	0.38	0.62	0.47	133

roc_auc _score: 0.9966523194643712

5. **All the evaluation results for each algorithm have been documented.**

6. **<u>Mention your final model, justify why u have chosen the same:</u>**

After evaluating the above algorithms on the CKD.csv dataset, we identified two best-performing models: Support Vector Classification (SVC) and Logistic Regression. Both models produced identical results across the evaluation metrics, including the F1 score, confusion matrix, classification report, and ROC-AUC score. Both algorithms achieved an accuracy, precision, and recall of 0.99, along with a ROC-AUC score of 1.00. Based on these evaluation results, we can select either of these two models for the next stage of the project: model deployment.